

BATSAMBUU BATBOLD

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EDUCATION

Macalester College | St. Paul, MN | Expected May 2027

B.A. Mathematics & Computer Science / GPA: **3.96/4.0**

Relevant Coursework: AI Robotics, Database Management Systems, Software Design & Development, Data Structures & Algorithms

United World College Maastricht | Netherlands | May 2023

IB Diploma (43/45), Bilingual Diploma

HONORS & AWARDS

- **Bronze Medalist, International Mathematical Olympiad** (2021): Placed top 250 globally in the World Championship Mathematics Competition for High School students
- **1st Place, Konhauser Problemfest** (2026): Won a multi-college math competition (Macalester, Carleton, St. Olaf, St. Thomas, Augsburg) — Macalester's first victory since 2018.
- **2× 1st Place, Macalester Local Programming Contest** (2025, 2026): Solved algorithmic competitive programming problems in C++.

EXPERIENCE

Full Stack Developer & Organizer | Young Mathematicians of Khuvsigul NGO | May 2019 – Present

- Built and deployed a full-stack [Mongolian math-olympiad platform](#) using Next.js, React, TypeScript, Supabase (PostgreSQL, Auth, Storage), and Tailwind CSS via Vercel.
- Architected row-level security with role-based access control, security-definer RPCs, and database-enforced moderation; wrote optimized SQL queries with indexing to power searchable problem archives, live contests, automated grading, and an admin CMS.
- Scaled the platform to 100+ registered users and 1,300+ daily visitors; organize free math camps and weekly online contests for 30+ rural students.

Mathematics Domain Expert (Frontier AI) | Mercor | Dec. 2025 – Jan. 2026

- Authored original competition-level math problems to evaluate state-of-the-art AI models; graded model-generated proofs and provided corrected solutions for AI training data.

Undergraduate Preceptor | Macalester College | Jan. 2024 – Present

- Support 30+ students per semester in Computational Linear Algebra and Probability; hold office hours and grade assignments.

RESEARCH & PUBLICATION

Research Assistant & Full Stack Developer | Macalester College | Summer 2026

- Analyzed a network of scientific concepts from 70 million academic papers across 150+ fields using Topological Data Analysis (TDA) to identify knowledge gaps and concept clusters.
- Built [TDA Cycle Explorer](#), a full-stack visualization app (FastAPI, React, Zustand, ECharts, Three.js) with RESTful API endpoints for exploring persistent homology cycles in large network data.
- Developed custom graph-layout algorithms for 10,000+ node networks, reducing cycle-edge crossings by ~15%.

[From Chaos to Continents](#) | Python, NumPy, Streamlit, Blender

- Engineered a procedural terrain generation engine using Voronoi Diagrams and Perlin Noise to create 50,000+ polygon worlds; implemented NumPy vectorization, reducing generation time by 40%.
- Accepted and presented at The International Symposium on Computational Geometry (SoCG 2026) as a multimedia submission.

PROJECTS

[Mosaic Photography Platform](#) | Next.js, TypeScript, PostgreSQL, Cloudflare, Vercel | Ongoing

- Developing a digital gallery portfolio builder with serverless image processing via Cloudflare R2 cloud storage, automated color palette extraction, EXIF metadata parsing, and on-demand lazy loading for high-resolution images.

[Art2Vec](#) | Python, Scikit-learn, NetworkX, React, TypeScript | May 2026

- Analyzed 16,000+ paintings as co-occurrence networks using Louvain community detection; built a k-nearest neighbors recommendation system and an interactive React web app with scrollytelling.

[NBA 2K Player Rating Simulator](#) | Python, Pandas, Scikit-learn, Streamlit | Dec. 2025

- Built a tool (200+ users) predicting NBA2K player ratings using XGBoost and Random Forest, achieving MAE of 8.58 across 36 attributes; constructed a data pipeline to scrape, clean, and merge NBA stats.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL, C++, Java, R, HTML/CSS

Frameworks & Tools: React, Next.js, Node.js, FastAPI, PostgreSQL, Supabase, Git, Vercel, Cloudflare, pytest, LaTeX, Claude Code, Codex

Technical Areas: Full Stack Development, Database Design, SQL Query Optimization, Cloud Deployment, Data Structures & Algorithms, Unit Testing